

Multi-Representational Reference Discovery for Progressive Retrieval Attention: Semantic, Lexical, and Iterative Routing Across Four Evidence Tasks

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August 22, 2026

Abstract

Progressive Retrieval Attention (PRA) separates a large logical memory from the smaller layer-native key/value working set admitted to attention. Contextual gists provide a semantic index, but names and token sequences can also act as addresses. We implement token-native exact, normalized, IDF-weighted, ordered, approximate, entity, and explicit channels as an identity-aligned sidecar to PRA’s bounded iterative router; decoded-word BM25 remains a separate baseline. The sidecar proposes identities; it neither copies nor materializes native K/V. At a matched four-chunk request budget over frozen Qwen3-0.6B features, QASPER favors exact matching, HotpotQA and 2Wiki favor BM25, and MuSiQue favors approximate token matching; static fusion never beats the best single channel. Splitting the budget into two root and two successor chunks changes the picture: root and successor representations differ on three of four datasets. True per-example root-channel headroom is 0.014–0.053 recall after separately accounting for validation instability, but neither a hand rule nor a validation-trained linear selector beats the best fixed policy. Rank fusion also fails. A competitive, gold-linked newly exposed address predicts positive iterative gain only on MuSiQue. Approximate matching repairs typos and some aliases, while every channel follows a confidently wrong reference. These are cohort-bounded discovery results: no K/V is materialized and no answer generation is run.

1 Introduction

PRA treats long context as addressable memory rather than a token sequence that must remain fully active. Its native-KV design separates stable object identity, compact routing representations, and detailed layer-specific K/V. This raises a distinct question: *what evidence should discover a candidate memory object?*

Current routing can score a contextual query $q_{t,\ell}$ against chunk gist $g_{i,\ell}$:

$$s_{i,\ell}^{sem} = \cos(q_{t,\ell}, g_{i,\ell}).$$

This captures contextual relatedness but need not preserve exact sequence identity. PRA, however, already has access to the host model tokenizer. Exact token spans, weighted token overlap, ordered token subsequences, and approximate subword matches preserve identity-bearing evidence without introducing a second tokenizer. They do not reliably resolve paraphrases or implicit relations, so contextual gists remain necessary. The four datasets below do not yield one winning formula.

They yield different lexical channels, substantial disagreement in the top-ranked chunk, and a per-example oracle gap. Retrieval channel is therefore itself a control decision.

Paper 2.5 studies bounded associative discovery and exposes entry-routing, granularity, broad-activation, and interference limits. This paper asks which discovery signals should generate and rank candidates. Paper 3 then asks which candidates, and how much detail, should be materialized. Keeping these questions separate is central to the experimental design.

2 Hybrid Reference Discovery

At hop h , let

$$\mathcal{C}_h = \mathcal{C}_h^{explicit} \cup \mathcal{C}_h^{tok-exact} \cup \mathcal{C}_h^{tok-approx} \cup \mathcal{C}_h^{semantic} \cup \mathcal{C}_h^{assoc}.$$

Candidates retain URI/chunk identity, layer, raw channel scores, discovery hop, parent, provenance, rank, confidence, and an explicit calibration-status flag. Exact handles may resolve deterministically; approximate channels generate candidates.

The implementation compares a progressive resolution cascade, candidate union, score fusion, token-native scoring followed by semantic reranking, semantic scoring followed by token-native reranking, and iterative hybrid traversal. Classical BM25 over decoded word terms is retained as a strong baseline rather than absorbed into the proposed mechanism.

3 Token-Native Reference Matching

The proposed non-neural resolver operates over the host model tokenizer. For each memory object we retain raw token IDs, normalized token spans, decoded subword strings, and title/name/alias boundaries. This permits exact identity matching without a second NLP vocabulary, while still allowing controlled normalization of tokenizer-specific leading-space, punctuation, and case artifacts.

A progressive cascade is evaluated:

explicit $\rightarrow$ exact token span $\rightarrow$ weighted/ordered token $\rightarrow$ approximate token $\rightarrow$ semantic gist.

Earlier high-confidence stages may resolve or sharply narrow candidates before later stages run.

For weighted overlap, let Q and R_i denote query and reference token multisets or aligned spans, and let $w(t)$ be a corpus-dependent token weight. A simple form is

$$s_i^{overlap} = \frac{\sum_{t \in \text{uniq}(Q) \cap \text{uniq}(R_i)} w(t)}{\sum_{t \in \text{uniq}(R_i)} w(t)}.$$

The present token-native score uses token-space inverse document frequency,

$$w(t) = \log \frac{N + 1}{df(t) + 1},$$

which can automatically downweight domain-generic tokens. Ordered subsequence overlap preserves multi-token identity while tolerating intervening material. Fixed stop-token removal and explicit token n -grams remain ablations.

Approximate matching is attempted only after exact or high-confidence matching fails or remains ambiguous. The current implementation uses sequence similarity over normalized subword pieces. Bounded token edit distance and similarity in the host model’s input-token embedding matrix remain open ablations rather than assumed defaults.

Classical word-level BM25 remains a strong baseline. The scientific question is whether model-token-aware identity resolution provides a better PRA-native discovery signal, not whether conventional information retrieval is unnecessary.

3.1 Confidence and Provenance

A candidate record includes raw channel scores and an explicit confidence field with stable intended semantics:

$$c_i \approx P(r_i \text{ is a valid referent} \mid q, \mathbf{s}_i, p_i),$$

where $\mathbf{s}_i$ contains channel-specific evidence and p_i records discovery provenance. The same numerical score from an exact token match, an approximate token match, and a semantic gist is not assumed to have the same reliability.

Raw discovery scores are marked uncalibrated in runtime traces. We fit calibration only on validation candidates and evaluate Brier score, expected calibration error, reliability, and selective accuracy/coverage. This makes the discovery interface testable for subsequent control:

$$c_i > \tau_{\text{act}} \Rightarrow \text{materialize},$$

$$\tau_{\text{search}} < c_i \leq \tau_{\text{act}} \Rightarrow \text{retrieve/disambiguate further},$$

with low-confidence cases eligible for abstention. External side-effecting action remains outside the scope of this paper, but the same confidence semantics can later support act/search/ask/abstain policies in agent systems.

4 Multi-Hop Reference Following

A useful pattern is

$$Q \xrightarrow{\text{semantic}} C_1 \xrightarrow{\text{token reference}} C_2 \xrightarrow{\text{reasoning}} A.$$

After selecting C_1 , its indexed token sequence becomes the lexical query for the next bounded hop. Its semantic gist remains the semantic frontier. Both channels therefore score the same aligned candidate rows before the existing branch, beam, depth, and unique-chunk controls admit C_2 . This produces a dynamically traversed chunk-reference graph without requiring a complete symbolic knowledge graph in advance.

If reference structure is informative, sparse traversal may replace broad semantic expansion:

$$\text{same evidence recall} \quad \text{with} \quad \text{fewer candidates/materialized tokens}.$$

5 Implementation

The implementation is intentionally split into a discovery sidecar and the inherited Paper 2.5 traversal. `TokenNativeIndex` in `src/prahf/hybrid.discovery.py`, beginning at line 161, tokenizes each source once and slices it with the cache’s stable token offsets. Its rows must match `GistIndex.records` exactly; a mismatch raises before routing. `DiscoveryCandidate` (line 100) retains URI, chunk and layer identity, all raw channel scores, spans, hop, parent, provenance, rank, and confidence. Longest exact spans and ordered overlap are implemented at lines 49 and 68. Channel scoring and policy selection begin at line 282.

Crucially, this code does not add a second traversal engine. `IterativeGistRouter.route` in `src/prahf/iterative.py` begins at line 413. Optional token arguments are validated at lines 432–441. Semantic and token-native scores are computed for each current frontier at lines 500–535; their fused ranking then passes through the original visited-set, per-parent branch, beam, confidence, and unique-chunk budget. An admitted chunk contributes its gist and token IDs to the next frontier. The retrieval graph records channel provenance, while native K/V remains untouched.

The product path exposes `token_iterative` and `hybrid_iterative` configuration modes. `PRAForCausalLM` constructs the aligned sidecar only for those modes (`src/pra_hf/model.py`, lines 428–445). After traversal stops, the existing identity mapper resolves selected chunks across consumption layers at line 449. Only after that mapping are graph nodes marked materialized (lines 457–461). Thus the experiment’s requested chunks are not mislabeled as active K/V.

The channel-geometry boundary is independently reusable. `src/pra_hf/channel_geometry.py` defines identity-set precision/recall and the deterministic channel oracle at lines 36–55, rare new-address detection at lines 90–106, and the gold-rejecting observable selector beginning at line 123. The experiment adapter in `experiments/paper2_6_hybrid_pra/run_channel_geometry.py` reconstructs one 32-token mean gist per frozen natural-feature span beginning at line 85, extracts root ranks, score gaps, lexicality, and evidence dispersion beginning at line 206, and starts the complete artifact-producing protocol at line 395. This split lets another implementation replace the diagnostic policy without changing candidate identity, traversal, or K/V materialization semantics.

The follow-up primitives are similarly explicit. Rank-only fusion, headroom decomposition, and the competitive-address predicate are at lines 63–122 of `src/pra_hf/channel_geometry.py`. The root/successor cross-product in `experiments/paper2_6_hybrid_pra/run_channel_selection.py` begins at line 216; the validation-only linear selector begins at line 386; extended collision controls begin at line 572; and the complete run begins at line 744. The machine-readable action contract is emitted near line 822 as `search_method_action_spec.json`. It names legal root and successor actions, parameters, confidence signals, costs, and failure indicators without importing gold geometry into the controller interface.

6 Research Questions and Success Levels

1. Does hybrid discovery improve supporting-evidence recall at matched active budget?
2. Which discovery signals dominate entry versus later hops?
3. Can iterative reference following outperform simply increasing semantic top- k ?
4. Do candidate provenance and confidence improve downstream materialization/control decisions?
5. When does token-level reference structure fail to add value over contextual gists?
6. Do retrieval improvements survive end-to-end answer generation?

We distinguish two success levels. A *discovery-level positive result* requires improved required-evidence recall at matched active cost, or matched recall with materially lower active memory/cost, supported statistically and by causal controls. An *end-to-end positive result* additionally requires a resolved improvement in downstream QA, such as paired EM/F1 improvement with an appropriate confidence interval or significance test on at least one natural multi-hop dataset.

This distinction is essential. Paper 2.6 studies discovery as a mechanism upstream of materialization and generation, so a robust discovery Pareto improvement remains scientifically meaningful even if the current model does not convert it into answer gains. However, the paper must not describe such a result as improved reasoning or QA performance.

7 Tokenization Robustness

Token-native matching introduces a specific invariance question: can the resolver recognize the same referent when its surface form induces a different tokenizer sequence? We therefore make

tokenization robustness a first-class experiment rather than a minor tokenizer-family ablation.

For a fixed referent r , construct controlled variants

$$\mathcal{P}(r) = \{r_{\text{case}}, r_{\text{space}}, r_{\text{punct}}, r_{\text{hyphen}}, r_{\text{split}}, r_{\text{typo}}, r_{\text{abbr}}, r_{\text{alias}}, \dots\}.$$

The present controlled suite covers case, punctuation, one-character misspellings, and contradictory high-overlap names. Whitespace, hyphenation, concatenation/splitting, naturally induced alternate subword segmentation, and abbreviation/alias tests remain open.

We measure degradation separately for exact-span, weighted/ordered, approximate-token, cascade, and iterative-hybrid conditions. The objective is an invariance curve showing how rapidly each mechanism loses referent accuracy as the surface form moves away from exact token identity.

8 Wrong-Reference Stress Testing

Average recall is insufficient if token-native matching becomes confidently wrong under ambiguity. We implement a controlled conflict in which an admitted chunk names both a high-overlap distractor and a lower-overlap valid target. Broader tests of shared aliases, nonexistent references, and independently varied semantic-token conflict remain open.

A particularly diagnostic condition is

$$s_{\text{token}}(A) \gg s_{\text{token}}(B), \quad s_{\text{semantic}}(B) \gg s_{\text{semantic}}(A),$$

with only one candidate being the valid referent. The present test measures wrong-candidate confidence and target recovery after the conflicting hop. Downstream QA degradation and broader selective behavior remain open. A useful hybrid method should degrade gracefully rather than convert ambiguous token evidence into high-confidence catastrophic routing errors.

9 Experimental Protocol

9.1 Controlled Mechanism Tasks

We construct 20 deterministic chains per condition. A semantic entry identifies the first chunk, which exposes a token address for a target placed behind two stronger semantic distractors. Clean, case, punctuation, typo, and confidently wrong-overlap variants use a two-chunk request budget. These are mechanism probes, not independent natural examples.

9.2 Natural Evidence Retrieval

We reuse the identity-disjoint Paper 2 feature protocol: frozen Qwen3-0.6B attention-input hidden states at layer 27, 32-token mean-gist chunks, revision `c1899de...e47ca`, and the question-exponential query. Validation contains eight HotpotQA and eight QASPER examples; held-out test contains 16 of each. No model weights are loaded in this study. B0–B4 and H1–H5 receive a final budget of four requested chunks. A1 deliberately buys breadth with eight semantic chunks and is not a matched-budget result. O1 admits annotated evidence first as a ceiling. No candidate is materialized.

The hybrid entry/later semantic weights are selected on validation from $\{0.4, 0.6, 0.8\} \times \{0.1, 0.3, 0.5\}$, with token weights as complements. Entry weights 0.4 and 0.6 tie in aggregate validation recall; the semantic-entry/token-later tie-break selects (0.6, 0.1). Test identities are

not used for selection. Confidence uses a monotone ten-bin histogram fitted only on validation candidates.

The extension reuses Paper 2.5’s separately frozen natural-graph cohort: 24 validation and 24 test 2Wiki examples, plus 18 validation and 18 test MuSiQue examples. Labelled evidence spans map through the fast tokenizer before the routing partition is constructed. Frozen token hidden states are averaged over the same 32-token local spans, yielding one gist per candidate chunk; the question-exponential query and layer remain unchanged. Six primary channels—gist, exact, BM25, approximate token, static token-to-semantic reranking, and two-hop iterative hybrid—all request exactly four unique chunks. Iteration uses branch two per step, depth two, at most four unique chunks, and reports all comparisons. The (0.6,0.1) entry/layer semantic weights are inherited without retuning. These cohorts remain below the requested 50 held-out units per dataset, so all new comparisons are explicitly cohort-bounded.

Per-example explanatory geometry includes query/evidence token overlap, IDF-weighted overlap, named-entity overlap, evidence regions, evidence extent and gaps, annotated depth, root ranks, score gaps, and channel disagreement. Gold-derived geometry is never passed to the adaptive baseline. Its fixed rule can see query rarity, root score gaps, channel disagreement, and whether an already admitted hop-one chunk exposes a rare address. The selector API rejects gold fields. Best fixed channels are selected on validation; the per-example oracle is evaluated only as an upper bound.

10 Baselines and Conditions

Condition	Discovery	Role
B0	gist-only PRA	semantic baseline
B1	BM25/word lexical	classical IR baseline
B2	exact token span/name	identity baseline
B3	weighted/ordered token	token-native baseline
B4	approximate token	fallback baseline
H1	semantic $\cup$ token-native	hybrid recall
H2	token-native $\rightarrow$ semantic	reranking
H3	semantic $\rightarrow$ token-native	reverse reranking
H4	token cascade + semantic fallback	cascade
H5	iterative token/semantic hybrid	main proposal
A1	broad semantic top- k	breadth baseline
O1	annotated/oracle	ceiling

Candidate budgets are matched for B0–B4 and H1–H5. A1 and O1 are labeled ceilings; no materialization budget is consumed in this discovery-only experiment.

11 Results

Table 1: Held-out evidence discovery. R, P, and M are supporting-evidence recall, precision, and MRR. Principal conditions request four chunks; A1 requests eight.

Condition	HotpotQA			QASPER		
	R	P	M	R	P	M
B0 gist	.151	.172	.276	.038	.062	.130
B1 BM25	.398	.453	.875	.042	.094	.135
B2 exact	.287	.328	.568	.178	.312	.443
B3 weighted	.287	.328	.568	.061	.109	.214
B4 approximate	.301	.359	.734	.037	.078	.141
H1 union	.209	.250	.365	.038	.062	.130
H2 token→semantic	.287	.344	.635	.044	.078	.125
H3 semantic→token	.279	.328	.635	.052	.094	.156
H4 cascade	.240	.281	.474	.126	.203	.307
H5 iterative hybrid	.279	.312	.646	.081	.125	.177
A1 broad semantic	.310	.180	.363	.088	.078	.155
O1 oracle	.876	1.000	1.000	.633	1.000	1.000

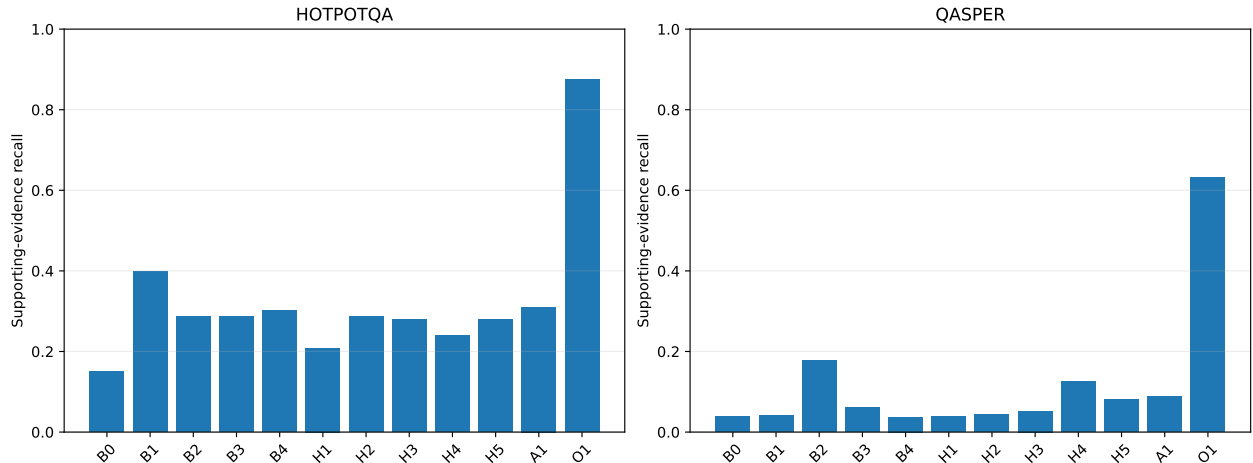


Figure 1: Held-out evidence recall. H5 improves gist-only recall on HotpotQA at the same four-chunk request budget, but does not surpass decoded-word BM25. QASPER favors exact spans. A1 uses twice the requested budget.

On HotpotQA, H5 improves recall over B0 by 0.129. A paired 10,000-sample bootstrap over the 16 test identities gives a 95% interval of [0.018, 0.242]; ten examples improve, four worsen, and two tie. B1 remains stronger, so the result supports heterogeneous discovery but not superiority of the present fusion rule. H5 reaches 90% of A1’s recall (0.279 versus 0.310) while requesting half as many chunks; this cross-budget comparison is descriptive.

On QASPER, H5 improves recall by 0.043, but its interval $[-0.014, 0.111]$ crosses zero (four improve, two worsen, ten tie). Exact spans instead improve over B0 by 0.140 with interval [0.053, 0.244]. This is evidence for regime dependence: QASPER’s scientific evidence in this small slice benefits from exact continuity, while repeated traversal adds little.

11.1 Four-Dataset Channel Geometry

Table 2 holds the request budget, chunk size, model, layer, and query representation fixed while changing only discovery channel. The winner changes across cohorts. QASPER favors exact spans; HotpotQA and 2Wiki favor decoded-word BM25; MuSiQue favors normalized approximate token matching, with static hybrid close behind. Static fusion improves over gist in every cohort but never beats the strongest single channel. The result is not “hybrid search wins.” It is that semantic and lexical representations fail differently, and their relative utility changes by example and task.

Table 2: Held-out recall/precision at four requested 32-token chunks. Bold marks the largest recall in each row; 95% paired-example bootstrap intervals for those winners appear below.

Dataset (n)	Gist	Exact	BM25	Approx.	Static hybrid	Iterative hybrid
QASPER (16)	.038/.062	.178 /.312	.042/.094	.037/.078	.044/.078	.081/.125
HotpotQA (16)	.151/.172	.287/.328	.398 /.453	.301/.359	.288/.344	.279/.312
2Wiki (24)	.132/.188	.196/.240	.262 /.312	.203/.250	.203/.250	.186/.250
MuSiQue (18)	.108/.333	.136/.417	.132/.431	.156 /.514	.151/.500	.146/.472

Best-channel recall intervals are QASPER [.096,.271], HotpotQA [.282,.516], 2Wiki [.176,.353], and MuSiQue [.122,.194].

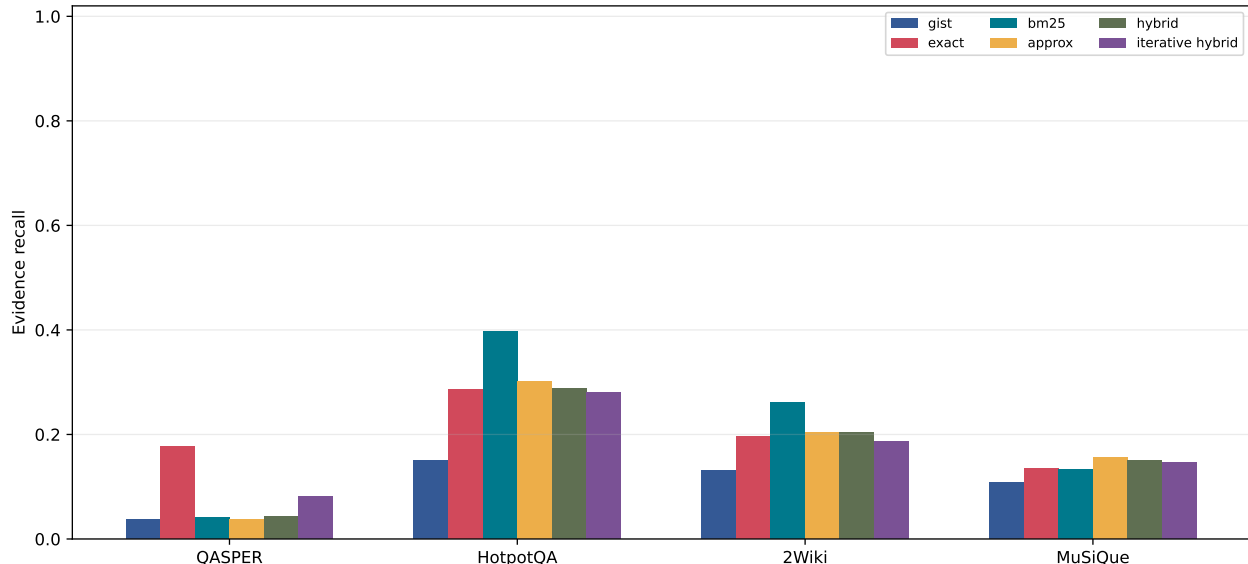


Figure 2: Matched-budget recall across retrieval channels and datasets. Different lexical channels win each cohort; neither static nor iterative hybrid dominates.

Precision moves with recall rather than revealing a hidden fusion win. The strongest precision is 0.312 for QASPER exact, 0.453 for Hotpot BM25, 0.312 for 2Wiki BM25, and 0.514 for MuSiQue approximate matching. Across all channel pairs, mean selected-set Jaccard is only 0.256 on QASPER, 0.325 on Hotpot, 0.369 on 2Wiki, and 0.326 on MuSiQue. In particular, gist–BM25 overlap is 0.045, 0.143, 0.152, and 0.119. The channels are not redundant, but naive fusion often spends its fixed budget on complementary distractors as well as complementary evidence.

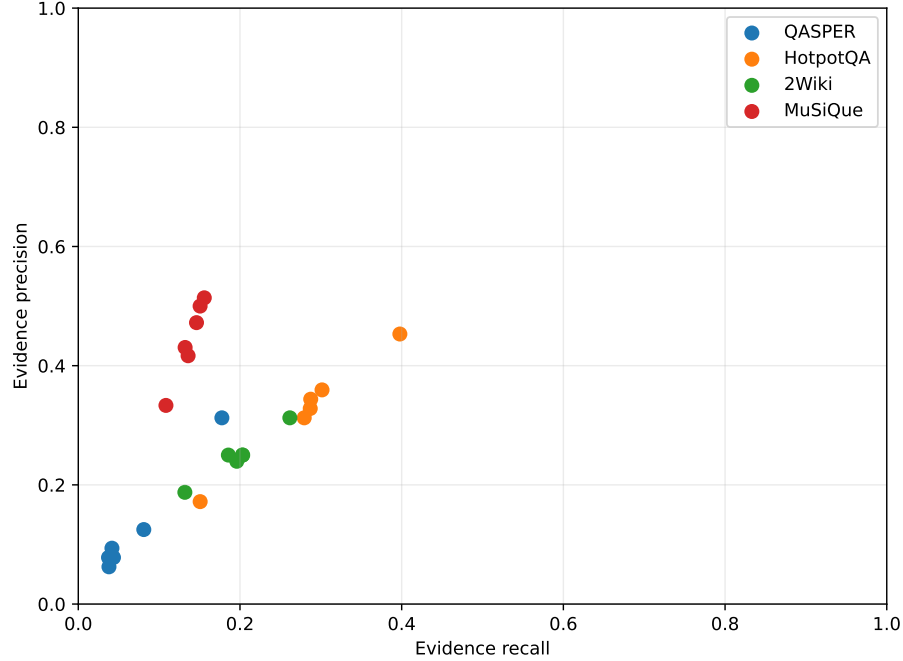


Figure 3: The channel precision–recall surface. Each point is one dataset/channel mean; higher recall obtained by fusion is not automatically a better discovery policy when precision falls.

Channel disagreement is itself diagnostic. Examples with at least the median three distinct root choices have a mean best-lexical-over-gist advantage of 0.197, compared with 0.101 below the median. High IDF-weighted query/evidence overlap similarly has a 0.200 lexical advantage, versus 0.131 in the lower half. These are post-hoc descriptive strata, not fitted dataset laws: the cohorts are too small and gold overlap is unavailable at deployment. Static hybrid remains below the best single channel in both strata (-0.124 and -0.082 recall), which argues against one uncalibrated score formula.

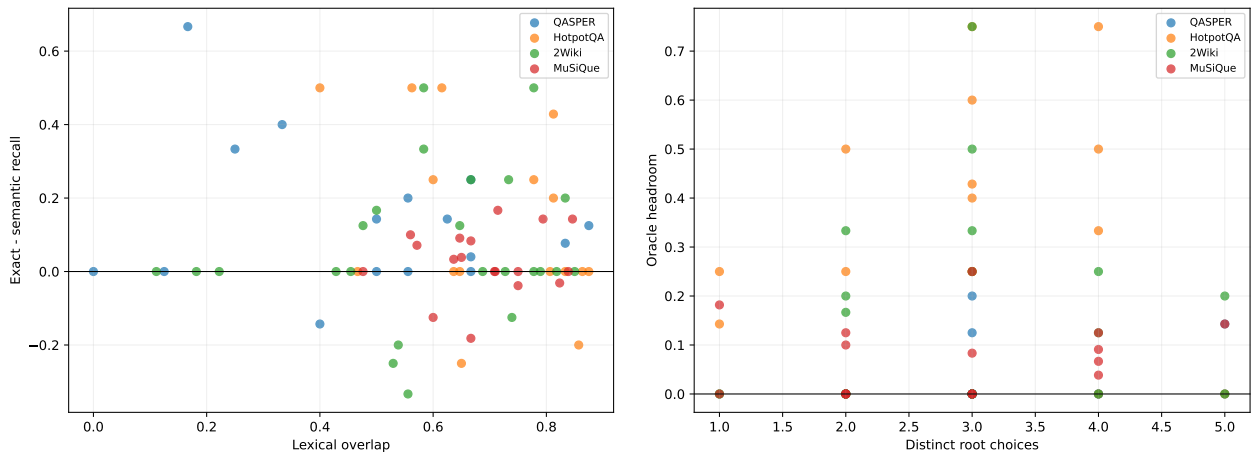


Figure 4: Left: gold query/evidence lexical overlap versus exact-over-gist recall. Right: root-channel disagreement versus validation-fixed channel-oracle headroom. Geometry is explanatory; neither variable is used as gold input to the selector.

11.2 Channel Selection and Iterative Exposure

The per-example channel oracle reaches recall 0.207 on QASPER, 0.489 on Hotpot, 0.315 on 2Wiki, and 0.199 on MuSiQue. Relative to the channel selected on validation, mean oracle headroom is 0.029, 0.338, 0.130, and 0.067. Hotpot’s large value partly reflects validation instability: gist beats BM25 by only 0.001 on eight validation examples, whereas BM25 is much stronger on test. On 2Wiki, validation selects iterative hybrid while test favors BM25. These reversals are evidence for larger calibration cohorts, not evidence that test labels should choose a production channel.

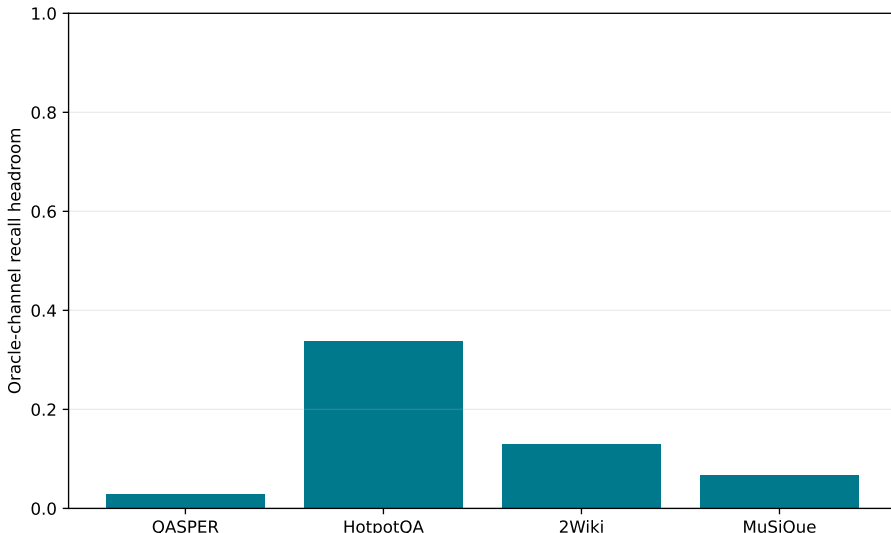


Figure 5: Per-example oracle headroom over the validation-selected fixed channel. The ceiling is useful for diagnosing adaptive opportunity, but is not a deployable result.

The fixed observable selector does not realize the oracle opportunity. Its recall is 0.081, 0.279, 0.186, and 0.146 across QASPER, Hotpot, 2Wiki, and MuSiQue. It therefore improves over static hybrid only on QASPER, ties the iterative policy elsewhere by construction, and remains below the best held-out single channel. We retain it as a leakage-audited baseline, not as the proposed final policy. A learned selector requires a larger validation cohort and should predict channel utility from query rarity, calibrated score gaps, and disagreement without using answer or evidence annotations.

A stricter new-address metric requires a normalized tokenizer piece to be absent from the question, present in an admitted hop-one chunk and unrecovered gold evidence, and rare under the example-local IDF. Such addresses occur in 6/16 QASPER, 5/16 Hotpot, 10/24 2Wiki, and 15/18 MuSiQue cases. Iterative-minus-static recall on address-positive cases is +0.100, −0.047, −0.005, and +0.003, respectively; address-negative effects are 0.000, +0.010, −0.027, and −0.042. Exposure is therefore necessary for the controlled lexical-following mechanism but not sufficient on natural data. The router must also identify the right address, preserve the right frontier, and rank the successor within the same total budget.

11.3 Separating Root and Successor Search

The preceding four-chunk result treats discovery as one decision. We next allocate two requested chunks to root search $Q \rightarrow C_1$ and two to successor search $(Q, C_1) \rightarrow C_2$. Every root channel is crossed with every successor channel for audit. The successor headline holds the root policy fixed to the validation-selected channel, so successor quality is not averaged over changing root quality.

For MuSiQue and 2Wiki, successor truth follows annotated dependency nodes; QASPER and Hotpot lack an ordered graph annotation, so successor truth is explicitly labelled as the remaining unordered evidence. This distinction prevents an inferred chain from becoming dataset truth.

Table 3: Best held-out root channel and best successor channel under the validation-selected root. Each stage requests two chunks. R/P are stage-local evidence recall/precision; comparisons are exhaustive candidate scores per example.

Dataset	Root	Root R/P	Root cmp.	Successor	Succ. R/P	Succ. cmp.	Same space?
QASPER	exact	.096/.312	143	exact new address	.025/.125	285	yes
HotpotQA	exact	.047/.094	50	native semantic	.135/.281	100	no
2Wiki	gist	.076/.125	42	exact new address	.050/.083	85	no
MuSiQue	approximate	.084/.139	82	native semantic	.032/.111	163	no

The best space changes after admission on three of four datasets. QASPER retains exact identity at both stages. Hotpot and MuSiQue prefer lexical root discovery but semantic successor expansion; 2Wiki prefers semantic entry and an exact newly exposed successor address. The low absolute recalls remain important: this is evidence that channel action should be state-dependent, not evidence that the present two-stage router solves multi-hop retrieval. Root choices also change with budget. At two root chunks, Hotpot favors exact rather than the four-chunk BM25 winner, showing that retrieval representation and conceptual budget interact.

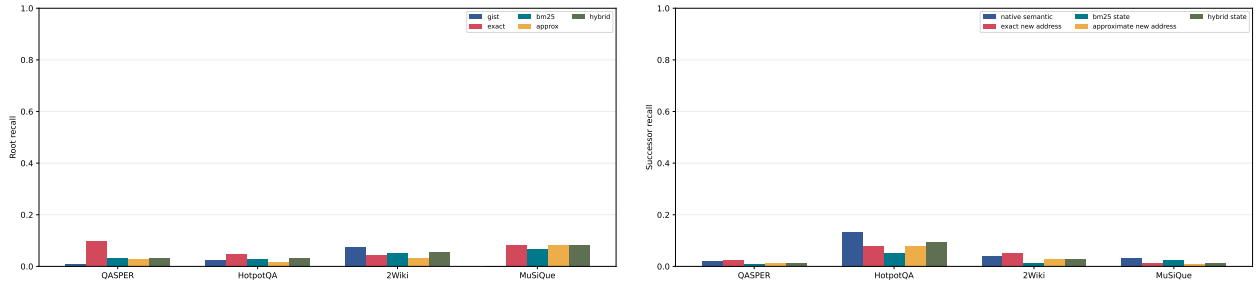


Figure 6: Root recall at two chunks (left) and successor recall at two chunks under the validation-selected root (right). Root and successor retrieval are different decisions even under one total four-chunk budget.

Per-example oracle transitions are heterogeneous. The most frequent are exact→exact (9), exact→native semantic (8), exact→BM25-state (7), gist→native semantic (6), and BM25→native semantic (5). No transition occupies a majority. The corresponding path gains are 0.038, 0.105, 0.076, 0.065, and 0.062. This is direct evidence against forcing root and successor states through one static fusion score.

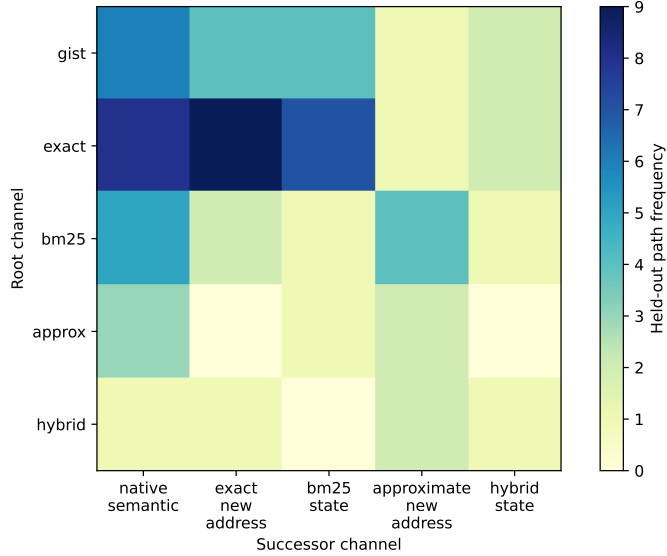


Figure 7: Held-out per-example oracle root→successor transitions. Frequencies describe an upper-bound diagnostic, not a deployable controller.

11.4 True Headroom and Observable Selection

We decompose oracle gain as

$$H_{selection} = R_{oracle} - R_{best\ heldout\ fixed}, \quad H_{validation} = R_{best\ heldout\ fixed} - R_{validation\ selected}.$$

Mean $H_{selection}$ is 0.014 on QASPER, 0.021 on Hotpot, 0.053 on 2Wiki, and 0.013 on MuSiQue. Mean $H_{validation}$ is 0, 0.031, 0.026, and 0. This corrects the earlier four-chunk “oracle headroom” figure, which combined adaptive opportunity with small-validation reversal. At the two-chunk root stage, 2Wiki has the largest true adaptive opportunity, while Hotpot’s validation instability is larger than its true selector headroom.

With the validation root held fixed, successor $H_{selection}$ is 0.027, 0.028, 0.056, and 0.016; successor $H_{validation}$ is 0, 0.056, 0.011, and 0.008. Thus 2Wiki retains the largest true channel opportunity at both stages, while Hotpot’s successor-channel validation reversal is twice its true adaptive headroom. The action space is real, but these small calibration splits cannot choose it reliably.

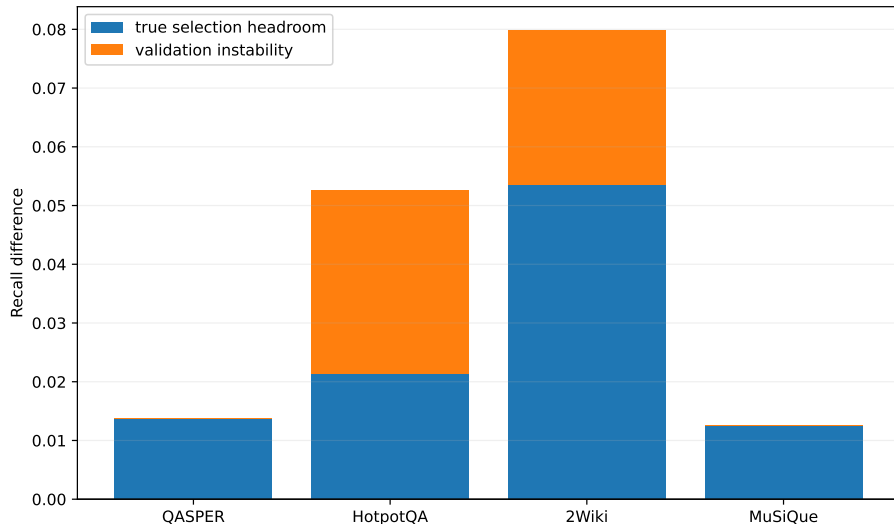


Figure 8: True per-example channel-selection headroom separated from validation-selected channel instability. Negative per-example instability is allowed; bars report means.

Neither modest selector baseline closes the gap. The fixed hand rule reaches root recall 0.020, 0.031, 0.040, and 0.073 across QASPER, Hotpot, 2Wiki, and MuSiQue. A validation-trained linear softmax selector over 24 query-only and score/rank features reaches 0.047, 0.016, 0.053, and 0.084. The best held-out fixed channels reach 0.096, 0.047, 0.076, and 0.084; the per-example oracle reaches 0.110, 0.068, 0.130, and 0.096. The linear model ties the fixed MuSiQue result and improves over the validation-selected 2Wiki channel by 0.003, but it does not justify replacing a fixed policy. No MLP is fitted because the 58 validation identities do not support a larger controller.

Root-channel disagreement also fails as a standalone selector feature. Its Pearson correlation with true per-example oracle headroom is only 0.052; selected-set Jaccard, top-10 rank overlap, and score disagreement have correlations 0.020, 0.107, and 0.013. Disagreement reveals representation diversity, but diversity alone does not say which channel is right.

11.5 Why Static Fusion Loses

At a two-chunk root budget, lexical channels add unique gold chunks relative to gist—0.812 per QASPER example, 0.188 Hotpot, 0.208 2Wiki, and 0.333 MuSiQue—but simultaneously add 3.062, 3.500, 2.792, and 2.778 unique distractors. Static hybrid recall is 0.031, 0.031, 0.053, and 0.082, below the per-example best single-channel recall of 0.110, 0.068, 0.130, and 0.087. The failure is therefore budget dilution plus score-scale mismatch, not lack of complementary evidence. Reciprocal-rank fusion avoids raw-score calibration assumptions but reaches only 0.035, 0.044, 0.053, and 0.045 recall. It is not a hidden robust winner.

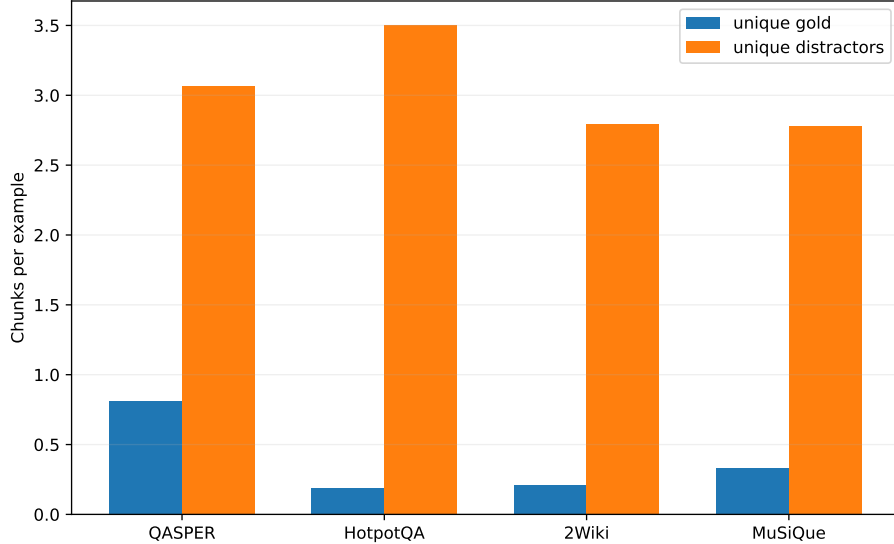


Figure 9: Unique lexical evidence and distractors relative to gist at the root stage. Complementary candidates are mostly distractors under the fixed two-chunk budget.

The inherited four-chunk MuSiQue analysis contains three cases where approximate retrieval strictly beats exact, BM25, and gist retrieval. All three have partial named-entity mentions under the observable taxonomy; this supports a partial-mention explanation on this tiny subset, not a general morphology or spelling claim. The two-chunk analysis finds no strict approximate-over-exact root win because the channels tie on the affected examples. QASPER’s three strict exact wins are classified as explicit evidence wording. The two strict Hotpot/2Wiki BM25 wins use multiple moderately informative terms rather than one dominant span. These manual-rule labels are auditable hypotheses for larger annotation, not ground truth causes.

11.6 Useful Addresses and Robustness

We tighten the iterative condition to

$$\text{UsefulAddress} = \text{Exposed} \wedge \text{GoldLinked} \wedge (\text{successor rank} \leq 4).$$

Under validation-selected roots, useful-address rates are 0.250 on QASPER, 0.063 on Hotpot, 0 on the 18 successor-bearing 2Wiki examples, and 0.222 on MuSiQue. UsefulAddress predicts a positive hybrid-state versus native-semantic path effect only on MuSiQue: +0.054 versus −0.001 without a useful address. QASPER is −0.021 in both strata; the single useful Hotpot case ties, and no 2Wiki address is competitive. Rare exposure is thus insufficient until the address also ranks within retry reach, and even the stricter predicate is dataset-dependent.

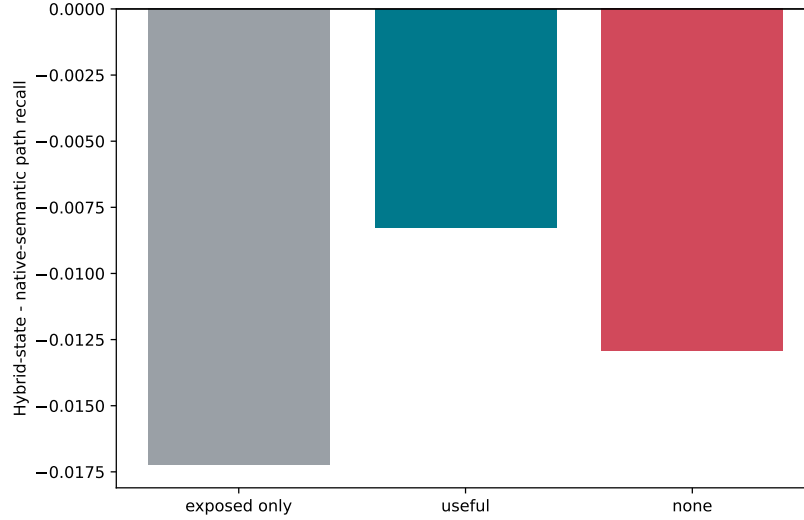


Figure 10: Path-recall gain of hybrid-state over native-semantic successor search. A competitive, gold-linked address is predictive only in the present MuSiQue subset.

The extended controlled suite separates normalization from disambiguation. All three channels recover clean, case, punctuation, near-entity, shared-prefix, numeric-ID, and URL/domain targets in all 20 trials without selecting the designated distractor. Exact matching recovers no typo targets and selects a near typo distractor in half; approximate matching recovers every typo and half of synonym/alias targets; iterative hybrid recovers half the typos but no aliases. Every channel follows the confidently wrong reference (target recovery 0, wrong-target recovery 1). Its hybrid wrong-target confidence is 0.396, below the diagnostic 0.5 action threshold, so retry is available in this constructed case, but calibration remains too weak for a deployment claim.

11.7 Controlled Chains and Failure Controls

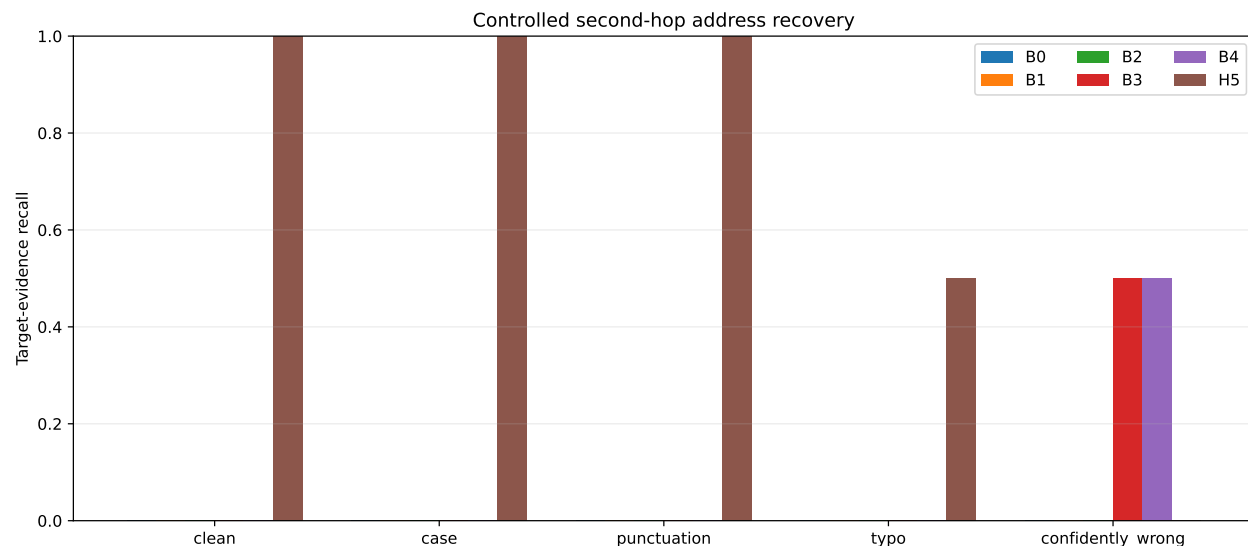


Figure 11: Controlled second-hop target recovery at a two-chunk request budget. H5 recovers every clean, case, and punctuation target, half of typo targets, and none when the exposed lexical address is confidently wrong.

Gist-only retrieval recovers none of the hidden targets because the target sits behind stronger semantic distractors. H5 recovers 100% of clean, case, and punctuation targets, showing that admitted evidence can expose a useful later-hop address. Recovery falls to 50% for one-character perturbations under the current approximate score. In the contradictory condition, H5 follows the wrong high-overlap address and recovers no targets; wrong selected candidates receive mean raw confidence 0.585. This is catastrophic rather than graceful degradation and prevents confidence-gated deployment without stronger disambiguation.

11.8 Confidence and Cost

Validation-fitted calibration remains weak. On held-out candidates, H5 has Brier score 0.185 and ECE 0.151, but no candidate crosses the 0.5 action threshold. BM25 has Brier 0.205 and ECE 0.096, and likewise has zero coverage at 0.5. Exact matching is particularly overconfident (Brier 0.281, ECE 0.251). Provenance is therefore useful diagnostically, but the present calibrator is not a production materialization controller.

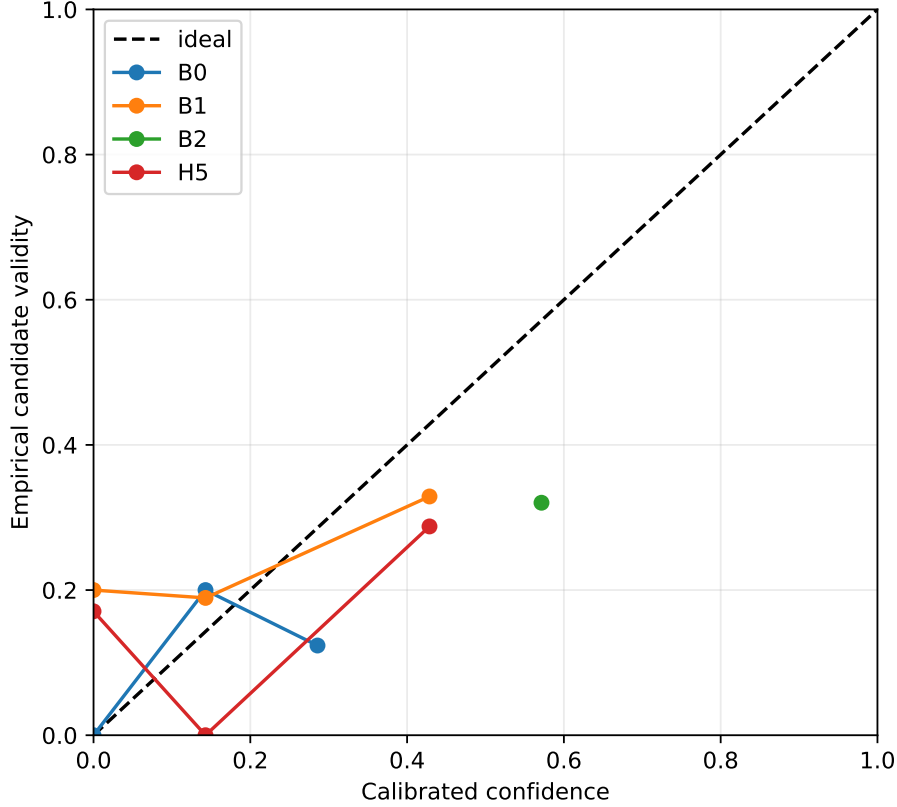


Figure 12: Held-out reliability after validation-only histogram calibration. Sparse and collapsed confidence levels limit threshold usefulness; these curves are diagnostics, not evidence of deployable calibration.

The implementation logs logical, resident, requested, and materialized counts separately. B0–H5 request exactly four chunks per natural example; materialized chunks and tokens are zero by construction. Averaged over the held-out examples, B0 takes 5.56 ms to reconstruct the gist index and 1.63 ms to route (7.19 ms cold total). B1 adds 34.42 ms of token-index construction and takes 62.60 ms to route (102.42 ms cold). Two-hop H5 adds 35.09 ms of token-index construction and takes 181.59 ms to route (224.46 ms cold), performing 289 mean semantic and 289 token comparisons versus B0’s 96 semantic comparisons. These are exhaustive Python research timings, not serving claims. Cached sidecars, posting lists, and tensorized lexical intersections are necessary before a systems comparison.

12 Evaluation

Discovery metrics include supporting-document/fact recall, MRR, NDCG, precision, missed-evidence rate, hop success, and path completion. Resource metrics include discovered, requested, and materialized chunks/tokens, active fraction, distractor materialization, index construction, cold/warm routing, repeated-hop cost, transfer, and end-to-end latency.

A useful descriptive efficiency quantity is

$$E_R = \frac{\text{required evidence recall}}{\text{materialized token-KV}},$$

but Pareto curves are primary because a single ratio can hide quality–cost tradeoffs.

An end-to-end extension would add answer EM/F1 and loss/calibration. The present retrieval headline explicitly records that answer generation was not run. Confidence metrics include Brier score, expected calibration error, reliability curves, and selective accuracy/coverage where feasible. The controlled wrong-reference and oracle conditions are reported here; disabled, shuffled, irrelevant, empty, and full-context generation controls remain for the downstream study.

13 Open Ablations and Controls

This study covers the complete policy matrix, four natural cohorts, matched primary budgets, raw and normalized exact spans, IDF-weighted/ordered overlap, approximate matching, per-example geometry, channel diversity, oracle headroom, a leakage-audited heuristic, perturbations, contradictory evidence, and provenance-aware calibration. It does not yet resolve stop-token suppression, token-embedding similarity, alias extraction, tokenizer-family sensitivity, chunk granularity, gist count, model size, sparse versus all PRA layers, or a statistically powered learned selector. Those are open ablations rather than implied completed experiments.

Metadata must not leak gold supporting labels. The exact query text/state used by each channel and hop is logged. Discovery is not counted as materialization, and hybrid systems do not receive an unreported larger active budget.

14 Interface to Materialization

The implementation exposes the discovery-side equivalent of

$$\text{discover}(q, M, B, \pi) \rightarrow \{(r_i, \mathbf{s}_i, p_i, c_i)\}_{i=1}^n,$$

where $\mathbf{s}_i$ preserves channel scores, p_i provenance, and c_i has the referent-validity semantics defined above. Paper 3 consumes these candidates and independently chooses an active native-KV set:

$$\text{materialize}(C, B_{KV}, \pi_M) \rightarrow \mathcal{A}.$$

Paper 3 can consume this record without making materialization depend on one retriever.

15 Contributions

This work contributes: (1) a provenance-preserving, multi-representational discovery formulation for PRA; (2) a model-token-native matcher evaluated beside decoded-word BM25; (3) one bounded iterative engine shared by semantic, token, and hybrid policies; (4) a four-dataset, matched-budget precision-recall and channel-diversity analysis; (5) a root/successor decomposition showing state-dependent channel transitions; (6) a separation of true per-example oracle headroom from validation instability; (7) leakage-audited hand-rule and linear selectors plus an RRF control; (8) a stricter competitive, gold-linked UsefulAddress diagnostic and extended collision controls; and (9) a machine-readable search-method action space for Paper 3.5 and a clean identity-level handoff to materialization.

16 Limitations and Falsification

The natural tests contain 16 QASPER, 16 HotpotQA, 24 2Wiki, and 18 MuSiQue examples. Only 18 held-out 2Wiki examples have a distinct annotated successor under this partition. The study

uses one frozen model, one routing layer, one tokenizer, one chunk size, and inherited feature captures made under two compatible but separately generated protocols. Bootstrap intervals quantify sampling uncertainty but do not substitute for larger replications. Validation contains only eight QASPER and Hotpot identities, which makes fixed-channel selection visibly unstable and the 24-feature linear selector underpowered. QASPER and Hotpot successor labels are unordered evidence remainders rather than annotated dependency edges. Geometry strata and win taxonomies are descriptive and partly gold-derived; they explain failures but cannot be deployment inputs. Token indexes add storage and runtime cost; the exhaustive Python scorer is too slow for serving claims. Most importantly, neither selector closes the oracle gap, UsefulAddress is predictive only on MuSiQue, and confidently wrong references defeat every channel.

The end-to-end claim has a separate bar. We did not run generation, EM/F1, or native-KV consumption in this study. The results therefore establish a discovery mechanism and identify a discovery–consumption question; they do not establish improved reasoning. QASPER and the wrong-reference control also show that a single fusion policy is not robust enough for deployment.

17 Conclusion

PRA discovery need not be one hidden-state similarity function. An identity-aligned token sidecar can participate in bounded gist traversal without touching native K/V. Across four small held-out cohorts, exact, BM25, approximate, and semantic channels each occupy a useful regime. Root and successor spaces differ on three datasets, and no oracle transition dominates. Static score fusion and rank fusion both dilute a tight budget with complementary distractors. Yet the observable selectors remain below the best fixed policy, and competitive addresses help only one cohort. The supported conclusion is therefore architectural rather than triumphant: retrieval representation is a state- and task-dependent action; Paper 3.5 should learn channel and effort jointly from larger calibration cohorts, while Paper 3 independently controls materialization. Until then, fixed policies, retry, and abstention remain safer than pretending one fusion rule is universal.